

Harness Pinout Worksheet

Stock LITTER-ROBOT Rev.2C control board — connector pinout as determined by direct multimeter continuity trace, physical hall sensor teardown, and bench testing with the diagnostic firmware. All 13 pins across both connectors are now resolved to a specific component and function.

Rev. 2026-07-05

Source: continuity trace + sensor teardown + bench test

Supersedes: fabacademy.org & robotshop paraphrases

13 / 13

connector pins with confirmed function

3

circuits fully resolved (motor, hall sensors, weight/anti-pinch)

0

physical rework remaining — hardware split done 2026-07-05

Connectors — as traced

Board-side pin → destination component

4-Pin Connector

MOTOR

- 1 **12VDC**
supply rail
- 2 **GND**
common ground
- 3 **Motor drive output**
→ U5 "43949SLBT" pin 7
- 4 **Motor drive output**
→ U5 "43949SLBT" pin 10

Reuse: pins 3 & 4 are the stock motor's two leads (driven, originally, by U5). These are the wires to land on the new L298N's OUT1/OUT2. Pins 1/2 are bulk 12V/GND — likely the main incoming

7-Pin Connector

HALL

WEIGHT

ANTI-PINCH

- 1 **GND**
common ground — hall sensors and both switches
 - 2 **Hall sensor "B" — OUT**
330Ω network → PIC16C622A pin 12
 - 3 **Hall sensor "A" — OUT**
330Ω network → PIC16C622A pin 10
 - 4 **Hall sensor "A" — VCC**
direct → PIC pin 14
 - 5 **Hall sensor "B" — VCC**
direct → PIC pin 14 (same net as pin 4)
- Weight switch**

power feed, exact downstream routing not traced further.

6

direct → PIC pin 6 – series-wired with pin 7, see callout

7

Anti-pinch switch

direct → PIC pin 5 – series-wired with pin 6, see callout

Hall sensors — fully resolved. Opening the sensor package found two Allegro A1101EUA-T ICs facing each other. Pins 4/5 (the "shared net" bridged on the PCB) are simply both sensors' VCC tied to one PIC-supplied power line — confirmed, not inferred, once the package was opened. Datasheet-confirmed: open-drain output (matches the 330Ω pull-up), VCC 3.8-24V, so 5V on pins 4/5 is correct. Only remaining question: which physical sensor (A/B) is Home vs. Dump.

Weight switch + anti-pinch switch — fully resolved, hardware split done (2026-07-05). These are two separate switches that the stock harness originally wired in **series on one shared loop** — confirmed by bench test and independently corroborated by a third-party teardown of this same board family. Read as one signal, that circuit couldn't tell "cat present" from "pinch detected," and a mid-cycle pinch produced zero signal change (no software fix exists for this — proven, not assumed). Fixed by opening the case, cutting the internal jumper connecting the two switches, and giving each its own ground return — now bench-verified reading fully independently. Firmware: PIN_ANTI_PINCH on GPIO14 (moved from GPIO13 2026-07-06 — the ESP32 mini board in use doesn't break out GPIO13), independent SAFETY_STOP interlock.

Polarity bug caught during verification: the anti-pinch switch is **normally-closed**, reading "on" at rest and "off" only when actually tripped — the opposite of every

other switch here. The firmware originally assumed uniform active-low polarity everywhere, which would have made the safety interlock trigger constantly during normal operation while missing a real pinch entirely. Fixed with a per-input `activeHigh` flag in `DebouncedInput`. See `CLAUDE.md` for the full story — worth remembering as a general lesson for any fail-safe (normally-closed) switch, not just this one.

New-board wiring worksheet

Fill in wire colors as you land each connection

STOCK PIN	FUNCTION	NEW BOARD TARGET	WIRE COLOR	LANDED
4-pin · 1	12VDC supply	12V in (buck conv. + L298N)	e.g. red	☐
4-pin · 2	GND	common GND	e.g. black	☐
4-pin · 3	Motor lead A	L298N OUT1	wire color	☐
4-pin · 4	Motor lead B	L298N OUT2	wire color	☐
7-pin · 1	GND	common GND	e.g. black	☐
7-pin · 6	Weight switch	GPI032 (INPUT_PULLUP)	wire color	☐
7-pin · 7	Anti-pinch switch	GPI014 (INPUT_PULLUP) – after physical switch split, see callout above	wire color	☐
7-pin · 2–5	Hall sensors (A1101EUA-T x2)	not reused – new/bench-mounted hall sensors used instead	–	–

Remaining work

All items closed out as of 2026-07-05

- ☒ Open the case and physically split the weight/anti-pinch switch loop. Done 2026-07-05 — internal jumper found and cut, both switches now have their own ground return.

- ✓ **Bench-verify both switches read independently.** Done 2026-07-05 — all three test cases (weight alone, anti-pinch alone, both together) confirmed correct with the diagnostic tool. Also caught and fixed a real polarity bug along the way — see the callout above.
- ✓ **Determine Home vs. Dump assignment.** Done 2026-07-05 — resolved by convention rather than by identifying the original board's A/B sensor labeling: PIN_HALL_HOME=GPIO34, PIN_HALL_DUMP=GPIO35, whichever sensor is physically mounted at each position gets wired to the correspondingly-named GPIO. The stock board's specific A/B identity was never actually load-bearing for this build.

Sources – confirmed: multimeter continuity trace + bench function labels (2026-07-03); physical hall sensor teardown + weight/anti-pinch bench testing (2026-07-04); hardware split + polarity bug fix (2026-07-05). Independently corroborated: a third-party teardown of this same board family at fabacademy.org/2020/labs/agrilab/students/florent-lemaire (hall-sensor color code and the weight/anti-pinch series-wiring problem both match). Discarded: two AI-summarized "pinouts" that were internally inconsistent and contradicted the fabacademy data. Full narrative lives in this project's CLAUDE.md, "Original harness pinout" and "Key decisions" sections.